Braynr Track

🎯 The Challenge

This track asks you to build agents for the *process* of learning, not the answer at the end. Before going into specifics, two ideas need to be sharp: what we are designing **against**, and what we are designing **toward**.

The Problem: Eduslop and the Performance Trap

Something strange has happened to learning. The most capable AI systems in history are now embedded in the daily workflows of students everywhere, and the evidence suggests that understanding is getting *worse*, not better.

The term **eduslop** names the phenomenon precisely: educational AI that prioritises outputs over outcomes. Students receive instant summaries, pre-generated answers, and algorithmic shortcuts that create the illusion of learning while systematically bypassing the cognitive processes that produce actual understanding.

The distinction that matters here is between **outputs** and **outcomes**. Outputs are the visible artifacts students submit: completed essays, finished assignments, passed tests. Outcomes are the intellectual capacities students actually build: genuine understanding, analytical reasoning, knowledge that transfers across contexts. *Outputs without outcomes. The work exists; the learning does not.*

Active Learning and Social Learning as Reference Frameworks

If eduslop names the failure mode, **active learning** names the alternative, and it is worth being precise about what that means, because the phrase is often used loosely.

Active learning, as cognitive science understands it, is not simply "engaging" or "interactive." It is the process by which learners actively construct meaning: formulating questions, generating summaries, making connections, testing retrieval, explaining ideas in their own terms. **These are not study techniques, they are the mechanisms of understanding itself.** What researchers call **desirable difficulties**, the productive friction of not-quite-getting-it, of having to try, fail, and reformulate, are not obstacles to overcome but the substrate from which durable knowledge is built.

Practically, active learning takes many forms:

- **Keywording**: semantic extraction requiring conceptual understanding
- **Question generation**: converting declarative knowledge into retrievable test items, a core principle of retrieval practice
- **Note-taking**: personal synthesis and annotation
- **Summary creation**: compression requiring comprehension and prioritization
- **Mapping**: leveraging dual coding of textual and visual information
- **Dialogic chat**: AI-scaffolded conversations that generate new understanding through iterative exchange
- **Spaced retrieval**: evidence-based spaced repetition
- **Flashcards**: discrete retrieval units that operationalize the testing effect through active recall of isolated concepts

But there is a second dimension that contemporary EdTech has almost entirely ignored: **learning is not a solo activity. It never was.** Real understanding is forged in dialogue, through the need to make your reasoning understandable to someone else, through the negotiation that happens when two people disagree about what a text means. **Knowledge is not transmitted; it is co-constructed.** These are not supplements to learning; in many cases they *are* the learning.

The two dimensions, **individual cognitive engagement** and **collaborative knowledge construction**, together define the target for any tool worth building.

What we want you to build

Build **agents that observe and support the *process* of learning**, not the answer at the end. The right intervention at the right moment. An alternative explanation when the same paragraph has been re-read four times. A peer connection when collaboration would help. A question that surfaces what the user *thinks* they know but doesn't.

Braynr is one concrete substrate to build against, and the resources we provide make that path the most direct. But Braynr is the *starting point*, not the constraint.

What binds this challenge is the frame above, the same frame Braynr is itself building toward. Teams can build on Braynr's surfaces, build adjacent to them, or build something that lives entirely outside Braynr's specific architecture. The cognitive thesis is what we evaluate, not which substrate you chose.

We are explicitly **not** looking for: another chatbot wrapper, another summariser, another flashcard generator, another "ask me anything about your notes." Those have been built. They optimise outputs. We want agents that produce outcomes.

The hard part of this challenge is not the engineering. It's the cognitive design. The teams that win this track will be the ones where someone is thinking deeply about *how humans actually learn*, and the agents reflect that thinking.

🧱 Braynr: Architecture Overview

What follows describes Braynr's product and architecture. Braynr is the reference substrate for this track, and the resources we provide are organised around it. You are not required to build on Braynr; the frame above is what binds the challenge. But if you do build on Braynr, this is the shape of what you are building on.

Braynr is a desktop application, Windows and Mac, for studying and managing what you learn. Users upload their materials (books, slides, notes, audio recordings, videos), and the app generates structured outputs from them: notes, flashcards, summaries, mind maps, transcriptions. An AI Tutor explains, simplifies, builds metaphors and analogies. A spaced-repetition algorithm drives long-term retention. A focus mode strips distractions. The whole thing is a multimodal study library with collections, tags, and search.

One piece of context matters for how you think about Braynr technically.

Braynr is a desktop application that runs locally on the user's machine. All user data (notes, highlights, questions, summaries, cognitive traces) is stored locally in Markdown (`.md`) format.

This has real design implications. The learning artifacts Braynr produces are plain text files on a local filesystem, readable by any tool, portable, inspectable. **That is not a constraint, it is an affordance. Keep it in mind.**

The Source-Centric Model

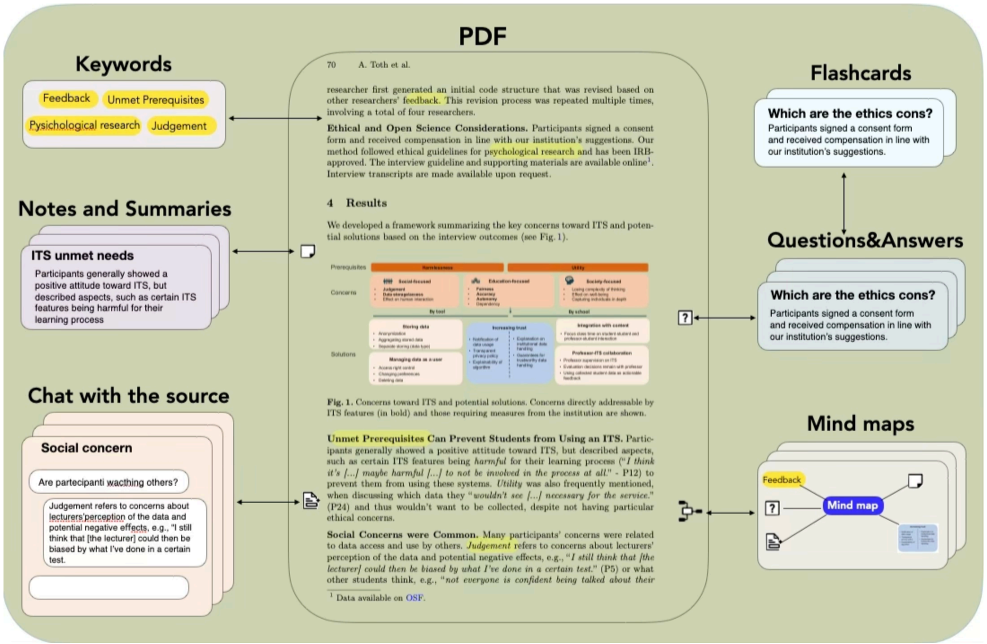

Figure 1: The source document as the centre of the learning architecture, with generative tools radiating from and bidirectionally linking back to specific positions in the source material.

Figure 1 captures Braynr's foundational principle: the source document is the centre of gravity of the learning experience. Everything else (keywords, questions, notes, summaries, concept maps, AI conversations, flashcards) radiates outward from it and links back to precise positions within it.

This matters for a reason that is more than aesthetic. When every generated artifact maintains a traceable connection to its origin in the source, **cognitive shortcuts become structurally useless.** You cannot fake a keyword that reveals genuine conceptual understanding. You cannot outsource a question that requires you to identify what you do not yet know. The architecture makes engagement *necessary*, not optional, because the only path through the material is the one that requires you to generate meaning from it.

The **bidirectionality** is the key feature. Not just links from source to elaboration, but backlinks from every elaboration to the specific passage that generated it. The learner can always trace what they produced back to where it came from. This is what closes the loop between source and understanding, and it is what most EdTech tools, which treat notes and source material as separate silos, never achieve.

The Three-Pillar Architecture

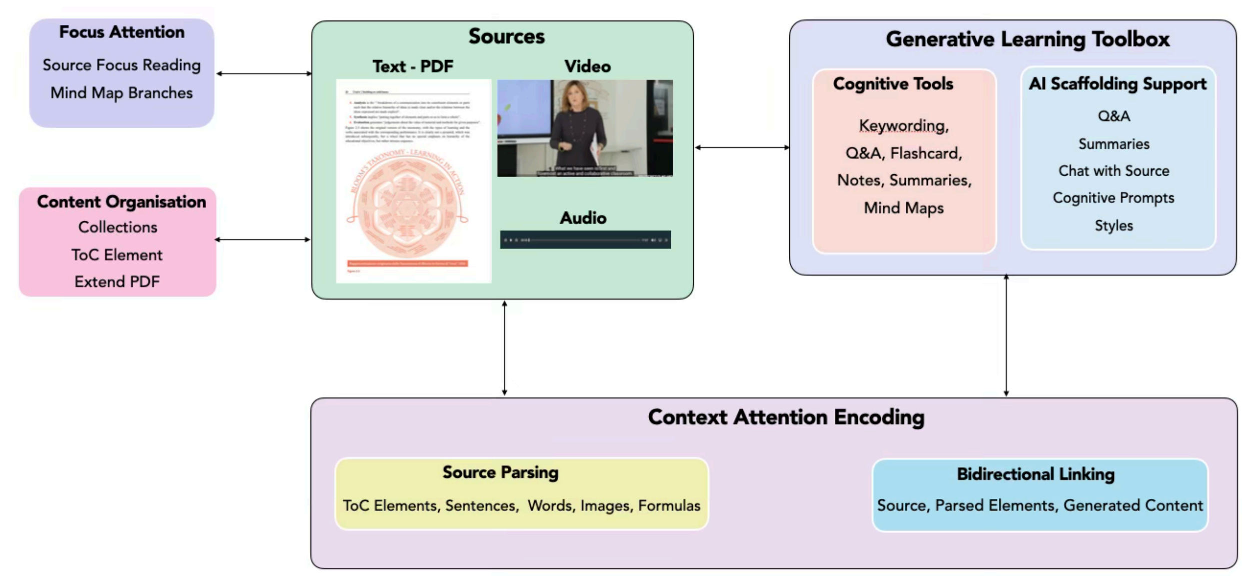

Figure 2: *The three interdependent pillars: Content Transformation, Pedagogically-Grounded Engagement, and Transformative Digital Technologies.*

Figure 2 shows how the broader architecture holds together. **Three pillars, none of which works without the others.**

- **Content Transformation** is the parsing layer: source material is decomposed into semantically addressable units: paragraphs, sentences, concepts, images, tables, each independently targetable. This is what makes granular engagement possible.
- **Pedagogically-Grounded Engagement** is where the learning actually happens. With content decomposed into addressable components, learners can engage through two complementary modes: *direct active generation* (producing keywords, questions, notes, maps, summaries) and *AI-assisted dialogic scaffolding*.
- **Transformative Digital Technologies**: parsing, bidirectional hyperlinking, local LLM scaffolding, are the enabling layer that makes the first two pillars possible at scale.

The Open Dimension

Braynr's current architecture is designed around the learner. The local-first, Markdown-based design produces **cognitive traces**: structured records of how a specific person moved through, annotated, and questioned a piece of source material, that accumulate over time.

What this means for builders

Five concrete facts about the technical surface you'll be building on:

- **Data lives on the user's machine.** Local-first. Files sit on the local filesystem.
- **Storage is markdown.** Every artifact Braynr generates (notes, summaries, questions, maps) is a `.md` file. Markdown is the native language of LLMs, so agent input is essentially free.
- **LLM calls go to cloud APIs.** Inference for the AI features goes out to Gemini, OpenAI, Claude, and similar. Your agent has the same freedom.
- **Cognitive traces are logged in JSON.** A structured event log captures user actions and their temporal sequence: clicks, reading time, prompt history, navigation. The goldmine for any agent that wants to understand *behaviour*, not just *content*.
- **Braynr Cards are the export format.** A Braynr Card aggregates a user's content, the conceptual links between concepts, and memory state, packaged as markdown. The cleanest, most structured input your agent can consume.

You don't need to build *inside* Braynr. Your agent can run anywhere (desktop, CLI, web) and consume Braynr Cards and cognitive traces as input. The data interface is the file system, and that is enough.

💡 The Design Space: Learning, Agents and Collaboration

Here is the tension worth sitting with.

Agent architectures that scaffold learning are becoming technically tractable. A two-layer model, in Braynr's case, generative learning tools plus observational and scaffolding agents, offers a concrete toolset for thinking about this. **Agents**

that detect struggle and offer scaffolding, promoting collaborations rather than answers for overcoming difficulties during the learning process, or opportunities for collaborations that create understanding and learning.

There is no prescribed answer here, that is precisely why this track exists.

The deeper question underneath all of this is not a UX problem: it is whether AI, in both its individual and social dimensions, **amplifies human cognitive agency or quietly displaces it.**

The Kind of Problem We're Talking About

What follows are three sketches of what an agent in this space might look like. They are not directions to take, sub-tracks to choose between, or a menu to pick from. They exist so you can develop a sense of the kind of problem that lives inside the frame, and recognize one when you find it. **Your job is to find your own problem inside the frame, not to extend any of the sketches below.** A team that builds something unanticipated, addressed seriously and grounded in the frame, will score better than several teams who all build variations on the same sketch.

A. Active Learning, *"Did I actually understand this?"*

A sketch where an agent observes a learner's engagement with the techniques of active learning and intervenes when the data suggests they need help.

- A user finishes a chapter. An agent could generate a question targeted at the most conceptually fragile part of what they just covered, not a random factual recall.
- A user re-reads the same paragraph four times. An agent could intervene with an alternative explanation, a worked example, or a referral to a different source rather than letting the loop continue.
- A user's answer to an assessment is shallow. Rather than saying "wrong," an agent could ask a follow-up that surfaces *why*, then offer scaffolding.

The hard part is not generating questions or alternative explanations. The hard part is choosing the *right moment* and the *right intervention*.

B. Learn to Learn, *"How does this person actually think?"*

A sketch where an agent treats prompt history and dialogue patterns as a window into a user's cognitive style, then helps them see and improve it. Whatever produces longitudinal signal becomes the substrate.

- A user whose prompts are mostly "explain this to me" could be shown their own pattern, with a gentle nudge toward more challenging engagement: comparison, critique, application.
- A user who initiates real dialogues (multi-turn, with pushback, with hypotheticals) might get a different kind of support than one who fires off isolated questions.
- An agent could build a longitudinal picture of *how this user learns best* across sessions and topics, and use it to shape future interactions.

Number, structure, and quality of prompts become metacognitive signals. An agent reflects them back to the user without judgment.

C. Collaborative Learning, *"How do others' processes inform mine?"*

A sketch where the *"knowledge is co-constructed"* principle becomes the architecture. An agent operates across multiple users and turns the group into a learning surface.

- Two users in the same study group are both stuck on the same concept. An agent could surface this, suggest a sync, or pull one user's notes (with permission) as a different angle for the other.
- A study group's collective notes and exports become a comparison surface: "three people in your group framed this concept differently; here are the contrasts."
- Peer-pattern matching: "users who worked through this material in your sequence found Concept X to be the bottleneck; here's what helped them."

The hard part is doing this in a way that respects privacy and doesn't degrade into social-media-for-studying. **Promote collaborations that create understanding, not noise.**

🛠️ What We Provide

From Braynr:

- A **premium Braynr account per team**, with all features unlocked.
- A **pre-loaded Braynr Card** with starter content, so your agent has clean structured input from minute one.
- **Direct mentor access** during the event from the Braynr team for product, architecture, and cognitive-design questions.

From GDG:

- **ElevenLabs**: voice synthesis credits.
- **Replit**: full project workspace.
- **Google AI Studio, Antigravity, Gemini CLI**: model access and tooling.

Anything else you want to use. The stack is yours. Bring your own keys for OpenAI, Anthropic, or any other provider if you prefer.

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A note on UX. The cognitive frame of this track is hard to land in 24 hours without a polished interface. The difference between an agent that *feels* like it understands the user and one that *technically* understands the user is almost entirely UX. Tools worth knowing about: Stitch for rapid UI prototyping, and Claude for design dialog and UX writing.

🏁 What to Deliver

All deliverables in **English**. Submit through the hackathon portal:

- **Demo video, 2 minutes max.** Upload to YouTube, submit the link.
- **Technical write-up, PDF, max 2 pages.** What you built, the cognitive design choices behind it, the architecture, what you'd do with another week.
- **Code repository.** Public GitHub or GitLab link. Include a README with setup instructions.
- **Presentation slides, PDF.** Used for your live demo to the jury.

⚠️ Mandatory Rules

- **Stay inside the cognitive frame.** Solutions that optimise outputs without addressing outcomes (answer accuracy, quiz performance, content summarisation) will be marked down regardless of technical polish. Re-read 🎯 The Challenge if unsure.
- **Live demo to the jury.** You will run your system live in front of judges. If something breaks live, your recorded demo video stands as evidence of working state, but live execution is the default expectation.
- **English for all deliverables.**
- **Bring-your-own everything is allowed.** Any data, any APIs (free or paid), any libraries, any model provider. Stay legal: no copyright violations, no scraped private data.
- **Originality.** The work must be yours. AI assistance for code, design, and writing is fully expected and encouraged, but the conceptual contribution and integration must be the team's.
- **Conduct.** GDG AI Hack 2026 code of conduct applies. Be excellent to each other.

🏆 Evaluation Criteria

Weight	Criterion	What judges evaluate
35%	**Cognitive Design & UX**	How clearly does the solution support the *process* of learning rather than measuring outputs? Does it build in desirable difficulties rather than smoothing them away? Is the user's experience intuitive, respectful, and growth-oriented? Did someone on the team think deeply about *how humans learn*, and does it show?
30%	**Innovation in Frame Adherence**	Does the solution interpret the cognitive frame in a non-obvious way? Did the team surface something interesting about Active Learning, Learn-to-Learn, or Collaborative Learning that wasn't anticipated? Bonus for combining directions or working at the

Weight	Criterion	What judges evaluate
		intersection of individual and collaborative dimensions.
20%	**Technical Feasibility & Architecture**	Does it work? Is the architecture sensible given a 24-hour build window? Where teams build on Braynr's surfaces, do they use them well rather than reinventing what is already there? Where teams build outside Braynr, is the substrate choice principled?
15%	**Demo & Communication**	Live demo quality, clarity of the elevator pitch and slide deck, ability to articulate the cognitive thesis behind the work. Can the team explain *why* their solution supports the process and not just the answer?

❓ FAQ

Can I bring my own data?

Yes. The Braynr Card you receive is a starting point so you don't waste time creating content. Use additional materials freely: your own notes, public datasets, anything legal.

Does my agent have to run locally?

No. LLM inference goes to cloud APIs in Braynr itself, so you have the same freedom. Local agents are welcome and you'll be technically aligned with Braynr's architecture, but not required.

What happens if Braynr breaks during the hackathon?

Mentors are on-site and reachable via the event channels. The Braynr Card you receive at kickoff are *your* files. It doesn't depend on Braynr being live.

What is the finalist follow-up?

The top teams in this track will have a conversation with Braynr's leadership immediately after the hackathon, about your work, your career interests, and potentially about working together. Braynr is hiring and is specifically interested in people who care about education and cognitive design.

Can my team work on multiple tracks?

No. Pick one main track at submission. Side challenges run independently in parallel.

🧭 Getting Started

1. **Read this brief twice.** The cognitive frame is the part that catches teams off guard. Make sure your team agrees on what "process of learning" means before you start coding.
2. **Install Braynr and use it for an hour.** Not as a developer, as a student. Take some real notes, generate some content, watch your own behaviour. The agent you build will be much better.
3. **Look at the Braynr Card and cognitive trace structure.** Understand the input shape your agent will work with. Sample provided at kickoff.
4. **Decide the team split.** This track *strongly* rewards a team that has at least one person doing pure cognitive design / UX thinking, not just engineering. If your team is all backend, find a way to compensate.
5. **Build something narrow and deep.** A single intervention that demonstrably works on real Braynr data beats a sprawling system that touches every direction at the surface.